

UCLA Math 131A, Winter 2025
Midterm 1, Lecture 3

Instructor: Linfeng Li

Monday, January 27

Please print your name and student ID below.

Name: _____

Student ID: _____

Please read all of the following rules before proceeding.

- The exam worth 40 points in total.
- Show all your work, and precisely state any theorems that you use.
- You are not allowed to use calculators or notes during the test, but you may use one-page, double-sided, and handwritten cheat sheet.
- Good luck!

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
Total:	40	

1. (10 points) Circle TRUE or FALSE for each of the following statement. No justification is needed.

- (i) True ☒ False For any $a \in \mathbb{R}$, if $\sqrt{a}$ is irrational then a is also irrational.
- (ii) ☒ True False The set $\{r \in \mathbb{Q} : 0 \leq r \leq \sqrt{2}\}$ has no maximum.
- (iii) ☒ True False Any finite set $S \subseteq \mathbb{R}$ has both supremum and infimum as real numbers.
- (iv) ☒ True False For any $a, b \in \mathbb{R}$, we have $|a - b| - |b| \leq |a|$.
- (v) ☒ True False The negation of the statement “ $\forall x \in \mathbb{R}$, if x is rational, then $2x^2 > x$ ” is “ $\exists x \in \mathbb{R}$ such that x is rational and $2x^2 \leq x$ ”.

2. (10 points) The factorial of $n \in \mathbb{N}$ is defined as $n! = 1 \cdot 2 \cdots n$. Use induction to prove that for any $n \in \mathbb{N}$,

$$\sum_{k=1}^n k \cdot k! = (n+1)! - 1.$$

Solution: Let P_n be the statement $\sum_{k=1}^n k \cdot k! = (n+1)! - 1$. The base P_1 is “ $1 \cdot 1! = 2! - 1$ ” which clearly holds. Now we assume P_n holds for some $n \in \mathbb{N}$ and we aim to prove P_{n+1} holds. We compute

$$\begin{aligned} \sum_{k=1}^{n+1} k \cdot k! &= \sum_{k=1}^n k \cdot k! + (n+1) \cdot (n+1)! \\ &= (n+1)! - 1 + (n+1)! \cdot (n+1) \\ &= (n+1)!(n+2) - 1 = (n+2)! - 1. \end{aligned}$$

Therefore, P_{n+1} holds. By induction, P_n holds for any $n \in \mathbb{N}$.

3. (a) (5 points) Suppose that $S \subseteq \mathbb{R}$ is non-empty and $\sup S = m_1$ is a real number. Prove that the supremum is unique, i.e., if there exists $m_2 \in \mathbb{R}$ such that m_2 is the least upper bound of S , then $m_1 = m_2$.
- (b) (5 points) Prove that $\sqrt{2 + \sqrt{2}}$ is irrational.

Solution: (a) Let $m_2 \in \mathbb{R}$ be the least upper bound of S . Since m_1 is an upper bound of S , we get $m_2 \leq m_1$. Similarly, $m_1 \leq m_2$ since m_1 is the least upper bound of S and m_2 is an upper bound of S . Hence, we conclude that $m_1 = m_2$.

(b) For the sake of contradiction, we assume that $\sqrt{2 + \sqrt{2}}$ is rational. By definition,

$$\sqrt{2 + \sqrt{2}} = \frac{p}{q}, \quad \text{where } p, q \in \mathbb{Z} \text{ and } q \neq 0.$$

Squaring both sides,

$$2 + \sqrt{2} = \frac{p^2}{q^2} \implies \sqrt{2} = \frac{p^2}{q^2} - 2 = \frac{p^2 - 2q^2}{q^2}.$$

Hence, we deduce that $\sqrt{2}$ is a rational number which is a contradiction. Hence, $\sqrt{2 + \sqrt{2}}$ is irrational.

4. Let $S \subseteq \mathbb{R}$ be non-empty and bounded from below. Let $c \in \mathbb{R}$. Define the set

$$E = \{x + c : x \in S\}, \quad \text{the collection of } x + c, \text{ where } x \in S.$$

- (a) (2 points) Prove that E has a greatest lower bound.
(b) (8 points) Prove that $\inf(E) = \inf(S) + c$.

Solution: (a) Since $S \subseteq \mathbb{R}$ is nonempty and bounded from below, we infer $\inf(S)$ exists as a real number by the completeness axiom. For any $x \in S$, $x + c \geq \inf(S) + c$. Hence, the set E is bounded from below by $\inf(S) + c$. It is clear that E is nonempty since S is nonempty. By completeness axiom, E has a greatest lower bound.

(b) We proved in (a) that $\inf(S) + c$ is a lower bound of E so $\inf(E) \geq \inf(S) + c$. On the other hand, for any $x \in S$, we have $x + c \geq \inf(E)$. It follows that $x \geq \inf(E) - c$, $\forall x \in S$. Thus, $\inf(E) - c$ is a lower bound of S and we get $\inf(S) \geq \inf(E) - c$. Therefore, $\inf(S) + c = \inf(E)$.